LAB SHEET 01: CORE WEB STRUCTURES USING SEMANTIC HTML

Task 1.1: Design a structured HTML resume page incorporating semantic layout blocks (<header>,<section>, <footer>) ensuring complete conformance with modern accessibility parameters. **Answer 1 — Task 1.1**

Structured HTML Resume Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Accessible HTML resume of a Computer Science student">
  <title>My Resume</title>
  <style>
    * {
      box-sizing: border-box;
    }
    body {  font-family:
Arial, sans-serif;  line-height:
1.6;  margin: 0;
background: #f4f4f4;  color:
#222;
    }
    header, main, footer {
max-width: 900px;  margin:
auto;
    }
    header {
background: #1f2937;
      color: white;
padding: 30px; text-align:
center;
```

```
        }
        main {
        background: white;
        padding: 25px;
        }
        section {
            margin-bottom: 25px;
        }
        h1, h2 {
margin-top: 0;
        }
        h2 {
            color: #1f2937;            border-
bottom: 2px solid #ddd;
padding-bottom: 5px;
        }
ul {
            padding-left: 25px;
        }
a {
            color: #005fcc;
        }
        a:focus,        a:hover {
outline: 2px solid #ff9800;
outline-offset: 2px;
      }
      footer {
          background: #1f2937;
          color: white;
           text-align: center;
        padding: 15px;
```

```
    }
  </style>
</head>
<body>
  <header>
    <h1>Happy Sharma</h1>
    <p>
      B.Tech Computer Science Student | Aspiring Software Engineer
    </p>
<address>
      Email:
      <a href="mailto:happysharma112007@email.com">
happysharma112007@email.com
      </a>
<br>
      Phone No:
      <a href="tel:+918923243391">
        +91 8923243391
      </a>
    </address>
  </header>
  <main>
    <section aria-labelledby="objective-heading">
      <h2 id="objective-heading">Career Objective</h2>
      <p>
        Motivated Computer Science student with knowledge of
programming, web development, and problem solving.  Seeking
opportunities to develop technical skills and  contribute to real-
world software projects.
      </p>
```

```
</section>
<section aria-labelledby="education-heading">
  <h2 id="education-heading">Education</h2>
  <article>
    <h3>B.Tech in Computer Science and Engineering</h3>
    <p>
      Coer University
      <br>
      2024 – 2028
    </p>
  </article>
  <article>
    <h3>Class 12th</h3>
    <p>MRPI College | 2024</p>
  </article>
</section>
<section aria-labelledby="skills-heading">
  <h2 id="skills-heading">Technical Skills</h2>
  <ul>
    <li>C++</li>
    <li>Python</li>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
    <li>SQL</li>
  </ul>
</section>
<section aria-labelledby="projects-heading">
  <h2 id="projects-heading">Projects</h2>
   <article>
     <h3>Online Shopping Website</h3>
```

```
            <p>
                Developed a responsive e-commerce website using
                HTML, CSS, and JavaScript with product pages,
shopping cart functionality, and search features.
            </p>
          </article>
          <article>
            <h3>Number Analysis Android App</h3>
            <p>
                Created an Android application that performs
operations such as prime checking, palindrome,
factorial, Armstrong number, and GCD/LCM.
            </p>
          </article>
        </section>
        <section aria-labelledby="experience-heading">
          <h2 id="experience-heading">Experience</h2>
          <article>
            <h3>Student Developer</h3>
            <p>
                Worked on academic programming and web development
projects while improving problem-solving and software
development skills.
            </p>
          </article>
      </section>
      <section aria-labelledby="achievements-heading">
          <h2 id="achievements-heading">Achievements</h2>
          <ul>
            <li>Completed multiple programming projects.</li>
            <li>Practiced C++ and Data Structures & Algorithms.</li>
```

```
            <li>Participated in technical learning activities.</li>
          </ul>
        </section>
        <section aria-labelledby="contact-heading">
          <h2 id="contact-heading">Contact & Links</h2>
          <p>
            LinkedIn:
            <a href="https://www.linkedin.com/in/happy-sharma-9a683a320"
              target="_blank"
rel="noopener noreferrer">
                LinkedIn Profile
            </a>
          </p>
          <p>
            GitHub:
            <a href="https://github.com/"
target="_blank"                rel="noopener
noreferrer">
                GitHub Profile
            </a>
          </p>
        </section>
    </main>
 </body>
</html>
```

Task 1.2: Construct an HTML page implementing a nested navigation menu bar using list item anchors, ensuring absolute path linking patterns to external resource addresses.

Answer 2 — Task 1.2

Nested Navigation Menu

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Navigation Menu</title>
</head>
<body>
<header>
  <h1>My Website</h1>
  <nav>
    <ul>
      <li><a href="https://www.google.com/">Home</a></li>
      <li><a href="https://www.example.com/">About</a></li>
      <li>Services
        <ul>
          <li>
            <a href="https://developer.mozilla.org/en-US/docs/Web/HTML">
              HTML
            </a>
          </li>
          <li>
            <a href="https://developer.mozilla.org/en-US/docs/Web/CSS">
              CSS
            </a>
          </li>
          <li>
            <a href="https://developer.mozilla.org/en-US/docs/Web/JavaScript">
              JavaScript
            </a>
          </li>
        </ul>
      </li>
```

```
        <li>Resources
          <ul>
            <li><a href="https://www.w3.org/">W3C</a></li>
            <li><a href="https://developer.mozilla.org/">MDN</a></li>
          </ul>
        </li>
        <li><a href="https://www.google.com/">Contact</a></li>
      </ul>
    </nav>
</header>
</body>
</html>
```

Task 1.3: Formulate an itemized web catalog interface containing descriptive metadata terms via structured definition lists (<dl>, <dt>, <dd>).

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Catalog</title>
</head>
<body>
<header>
    <h1>Product Catalog</h1>
</header>
<main>
    <section>
      <h2>Laptop</h2>
      <dl>
        <dt>Brand</dt>
```

```
        <dd>ABC Computers</dd>
        <dt>Category</dt>
        <dd>Electronics</dd>
        <dt>Processor</dt>
        <dd>Intel Core i5</dd>
        <dt>RAM</dt>
        <dd>16 GB</dd>
        <dt>Storage</dt>
        <dd>512 GB SSD</dd>
        <dt>Price</dt>
        <dd>₹55,000</dd>
        <dt>Description</dt>
        <dd>
            A high-performance laptop suitable for students
and programming.
        </dd>
      </dl>
   </section>
</main>
<footer>
   <p>© 2026 Product Catalog</p>
</footer>
</body>
</html>
```

Outout 1

Happy Sharma

B.Tech Computer Science Student | Aspiring Software Engineer

Email: happysharma112007@email.com Phone No: +91 8923243391

Career Objective

Motivated Computer Science student with knowledge of programming, web development, and problem solving. Seeking opportunities to develop technical skills and contribute to real-world software projects.

Education

B.Tech in Computer Science and Engineering

Coer University 2024- 2028

Class 12th

MRPI College | 2024

Technical Skills

- C++
- Python
- HTML
- CSS
- JavaScript
- SQL

Projects

Online Shopping Website

Developed a responsive e-commerce website using HTML, CSS, and JavaScript with product pages, shopping cart functionality, and search features.

Number Analysis Android App

Created an Android application that performs operations such as prime checking, palindrome, factorial, Armstrong number, and GCD/LCM.

Student Developer

Worked on academic programming and web development projects while improving problemsolving and software development skills.

Achievements

- Completed multiple programming projects.
- Practiced C++ and Data Structures & Algorithms.
- Participated in technical learning activities.

Contact & Links

LinkedIn: LinkedIn Profile

GitHub: GitHub Profile

Output 2 My Website

- Home
- About
- Services
  - HTML
  - CSS
  - JavaScript
- Resources
- W3C
  - MDN
- Contact

Output 3

Product Catalog
Laptop
Brand: ABC Computers
Category: Electronics
Processor: Intel Core i5
RAM: 16 GB
Storage: 512 GB SSD
Price: ₹55,000
Description: A high-performance laptop suitable for students and programming.